

Recognizing Palm Vein in Smartphones Using RGB Images

Shi-Jinn Horng , Dinh-Trung Vu , Thi-Van Nguyen , Wanlei Zhou , *Senior Member, IEEE*,
and Chin-Teng Lin , *Fellow, IEEE*

Abstract—Recently, demand for biometric access controls and online payments in smartphones increased, necessitating further investigation and development in this area. This article proposes a new low-cost palm vein recognition system for smartphones using red, green, blue (RGB) images. First, we detect and enhance palm vein patterns, using the saturation channel instead of the red channel as in the existing approaches. Then, to address the challenging contactless capturing problems of smartphones—such as scale variants, rotation, closed fingers, or rings on hand—we introduce an improved method for the region of interest extraction, based on the convex hull, with a new idea for key vector use. We also designed a new lightweight deep learning-based model for smartphones, which was overlooked in previous palm vein recognition studies. The proposed model comprises suitable blocks of convolution, depthwise separable convolution, inverted residual bottleneck, and spatial pyramid pooling module; in addition, the accuracy is enhanced with fusion strategy. Results show that the proposed model is both smaller and more accurate than related models. The integrated proposed model obtains the best equal error rate, 0.49%, and an inference time of 8 ms.

Index Terms—Deep learning, key vector, mobile palm recognition, saturation channel, spatial pyramid pooling.

I. INTRODUCTION

BIOMETRICS recognition systems use physiological and behavioral traits to identify individuals. Scientists in the physiological traits field have studied facial, retinal, fingerprint, and palmprint recognition systems extensively for years [1], [2]. The most commonly used biometric traits are fingerprints and faces; however, external biometrics, like fingerprints and faces, can be faked using stolen or replicated images. Moreover, aging, dermatitis, accidents, and other factors affect these traits. In contrast, palm vein recognition systems address these limitations, and therefore, draw significant attention in the industry. Palm veins are blood vessels underneath the palm skin and are unique to an individual. The veins remain stable over time. Therefore, it is extremely difficult to copy or forge vein patterns.

Scientists have studied palm vein recognition systems for both theoretical to practical applications. Two types exist: hand-crafted [3], [4], and deep learning [6]–[8]. In practical applications, the palm vein identification system has been developed commercially, such as Fujitsu PalmSecure™, US 2005/0148876 A1, LG G8 ThinQ. However, the details of the design and architecture of practical applications remain confidential, due to trade secrets. In addition, both research and application focused on implementing specialized devices with near infrared (NIR) sensors, incurring additional costs.

Today, demand for biometric access controls and online payments in smartphones increased, due to security concerns in these devices. To date, our research team has not found any studies on palm vein systems for mobile phones. Moreover, we discovered the following disadvantages of conventional palm vein recognition research. 1) Prolonged exposure to the infrared radiation energy used to capture vein patterns is potentially harmful. 2) Integrating NIR sensors incurs additional costs. Motivated by the above limitations, we designed a new, lightweight palm vein recognition system for smartphones without requiring NIR sensors.

Palm vein recognition system for smartphones must locate the correct region of interest (ROI), extract and enhance the efficient palm vein features, and classify and verify individuals. Because of contactless capturing, challenging problems emerge, including scale variants, rotation, closed fingers, or rings on a

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Shi-Jinn Horng is with the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei 10607, Taiwan, and also with the School of Computer Science, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: horngsj@yahoo.com.tw).

Dinh-Trung Vu and Thi-Van Nguyen are with the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei 10607, Taiwan, and also with the Faculty of Information Technology, Vietnam Maritime University, Haiphong 04000, Vietnam (e-mail: trungvd@vmaru.edu.vn; vannt.cnt@vmaru.edu.vn).

Wanlei Zhou is with the School of Computer Science, University of Technology Sydney, Sydney, NSW 2007, Australia, and also with the Institute of Data Science, City University of Macau, Taipa, Macau 999078, China (e-mail: wanlei.zhou@uts.edu.au, wlzhou@cityu.mo).

Chin-Teng Lin is with the Computational Intelligence and Brain Computer Interface Lab (CIBCI) AAIL FEIT, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: chin-teng.lin@uts.edu.au).

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hand. The existing ROI extraction method based on hand shape [4] failed in large rotations. The methods using reference points [13] failed if the captured images had closed fingers and rings on the hand. In this article, we present an improved ROI extraction method addressing the above problems. We tracked the changes in the distance of contour points to the corresponding convex hull edge to remove unintentional valley edges resulting from closed fingers or rings. The novel key vector idea determines the hand direction and helps locate the correct ROI on any image rotation.

To detect and extract palm vein features, Cho *et al.* [3] claimed that palm vein patterns can be extracted from the red channel of red, green, blue (RGB) images, as this channel is close to the NIR spectrum. Conversely, we found that the palm vein lines have high contrast with other structures in the background and appear more prominently in the saturation channel than the red channel. Cho *et al.* [3] utilized a hand-crafted method for the palm vein verification system and ignored the investigation of smartphones. In the industry, deep learning outperforms hand-crafted learning, because of its ability to extract hidden knowledge from massive data with high variance. The deep learning method is more effective for open-set scenarios, when the images for testing differ from the training phase data. In this article, we design a deep learning model, using saturation channel images as an input for a palm vein recognition system. Several deep learning models have extracted features from NIR images. Zhang *et al.* [6] built a very large model PalmR_{CNN}, based on deep architecture “Inception_ResNet_v1” with large parameters. Thapar *et al.* [8] used a deep framework encoder-decoder to learn generative domain-specific features, and then, generated palm vein features. These two large models are not suitable for mobile phone applications because of their size. Zhong *et al.* [7] modified the popular convolutional neural network-F (CNN-F), but only evaluated it on a trained database. The CNN-F model’s accuracy decreased significantly, when testing with a nontrained database. In this study, we are the first to develop a contactless palm vein recognition system, using a deep learning-based method for mobile phones. The proposed model is called the mobile palm vein extraction network using saturation channel (MPSNet), which is composed of suitable blocks of convolution, depthwise convolution, inverted residual bottleneck, and spatial pyramid pooling module; then, the final accuracy is enhanced with fusion strategy.

In general, the differences and contributions between our approach and other methods are summarized as follows.

- 1) Palm vein detection: We extract and enhance palm vein patterns using the saturation channel from RGB images taken by regular smartphone cameras. The veins extracted from the saturation channel show higher contrast with the background than those extracted from the red channel in Cho *et al.* [3]. The proposed palm vein system recognition rate, using the saturation channel, proves to be better than using the red channel (see Table IV).
- 2) ROI extraction: Existing ROI extraction methods, such as [4] and [13], failed in large rotation, partial hand image captures, fingers closed, or wearing rings scenarios. We propose an improved palm ROI method based on a

TABLE I
EER (%) OF THE PROPOSED MODEL TRAINED WITH DIFFERENT LOSSES (OPEN-SET SCENARIO)

Method	Tongji	PolyU	CASIA	XJTU-UP	NTUST-HP	NTUST-IP
Softmax	3.15	1.93	7.30	4.08	1.85	16.46
L2-Softmax [26]	2.14	1.47	4.60	2.05	1.54	15.63
CosFace [27]	0.57	0.53	1.78	1.03	0.54	4.08
ArcFace [28]	0.53	0.50	1.71	0.99	0.51	4.04
AdaCos [21]	0.46	0.47	1.52	0.90	0.49	3.62

TABLE II
MODEL PARAMETERS AND SIZE (KERAS VERSION)

Method	Params (M)	Model Size (MB)
MobileNetV1 [9]	3.36	13.06
MobileNetV2 [17]	2.42	9.69
MobileNetV3 [18]	3.11	12.38
MobileFaceNet [20]	1.03	4.41
MPSNet (Ours)	0.59	2.54

TABLE III
EER (%) COMPARISONS OF VARIOUS DL-BASED METHODS ON NIR DATABASES

Method	Tongji		PolyU		CASIA	
	CS	OS	CS	OS	CS	OS
PalmR _{CNN} [6]	N/A	2.30	N/A	N/A	N/A	N/A
DHPN [7]	N/A	N/A	0.0222	N/A	N/A	N/A
PVSNet [8]	N/A	N/A	0.6600	N/A	3.71	11.44
MobileNetV1 [9]	0.0131	1.37	0.0269	1.91	0.65	3.50
MobileNetV2 [17]	0.0283	1.34	0.0410	1.98	0.72	3.60
MobileNetV3 [18]	0.0212	1.19	0.0313	2.02	0.59	3.53
MobileFaceNet [20]	0.0015	0.86	0.0046	0.77	0.41	2.62
MPSNet [Ours]	0.0001	0.46	0.0011	0.47	0.17	1.52

TABLE IV
EER (%) COMPARISONS OF VARIOUS DL-BASED METHODS ON RGB DATABASES (OPEN-SET SCENARIO)

Method	XJTU-UP		NTUST-HP		NTUST-IP	
	Red	Sat	Red	Sat	Red	Sat
MobileNetV1 [9]	1.78	1.66	1.32	1.20	12.91	11.77
MobileNetV2 [17]	2.03	1.98	1.41	1.33	11.81	11.39
MobileNetV3 [18]	2.11	1.89	1.47	1.42	11.05	10.39
MobileFaceNet [20]	1.34	1.28	0.62	0.57	8.89	7.97
MPSNet [Ours]	1.08	0.90	0.55	0.49	4.68	3.62

Notes: Sat = Saturation channel.

convex hull with a new key vectors idea that solves the aforementioned problems.

- 3) Feature extractor: We propose a new, lightweight, deep learning-based network, MPSNet, for extracting palm vein features from an input saturation channel, specialized for smartphones. Comprehensive experiments conducted on several public and self-collected databases show that the proposed model is both smaller in size and more accurate than previous related methods. Our proposed recognition system, implemented on two smartphones in a real-time environment, achieves high inference speed and accuracy.

Algorithm 1: Extracting Saturation Channel

Input: RGB image;
Output: saturation channel;
 $R=R/255$, $G=G/255$, $B=B/255$;
 $V_{\max}$ =maximum of R , G , B ; $V_{\min}$ =minimum of R , G , B ;
 $L=(V_{\max}+V_{\min})/2$;
if $L<0.5$ **then** $S=(V_{\max}-V_{\min})/(V_{\max}+V_{\min})$;
else $S=(V_{\max}-V_{\min})/(2-(V_{\max}+V_{\min}))$;
end if
 $S=255\times S$;

- 4) A self-collected, contactless database consists of 7940 RGB images. The source code will be shared for research purposes at.¹

The rest of this article is organized as follows. Section II describes the background knowledge related to the contactless palm vein recognition system. Section III presents the details of the proposed method and its applications. Section IV presents the experimental results and discussion. Finally, Section V concludes this article.

II. RELATED WORK

In this section, we present several recent deep-learning approaches for palm vein recognition. Each method was performed and evaluated using the NIR databases. PalmR_{CNN} [6] used self-collected palm vein images to fine-tune a large model, modified from “Inception_ResNet_v1” trained on palmprint images. Then, the trained model extracted the palm vein features, and the Euclidean distance was used in the verification phase. The equal error rate (EER) of PalmR_{CNN} [6] was 2.3% in the Tongji database. Palm vein Siamese network (PVSNet) [8] used an encoder-decoder network to learn generative domain-specific features, followed by the generation of palm vein features. Researchers applied triplet loss for training, and the L2 distance was used for matching the templates. Usually, the EER of training and testing with the same database is significantly lower than that of different databases. Using the PolyU database for training and testing, PVSNet [8] obtained an EER of 0.66%, but EER increased to 11.44%, if the training phase was performed on the PolyU database and testing on the CASIA database. Still, improvement in accuracy rates remains a need. Moreover, the sizes of these two very large models make them unsuitable for mobile phone applications.

Deep hashing palm network (DHPN) [7] was modified from the popular CNN-F architecture to extract features. The extracted features are represented in binary codes, and the matching step uses the Hamming distance. Similarly, DHPN [7] obtained an EER of 0.0222% using the PolyU database for training and testing, but the EER increased to 1.33%, if tested using their self-built palm vein NIR images obtained from 60 people. Obviously, deep learning-based methods are typically developed to extract features from NIR databases. No current

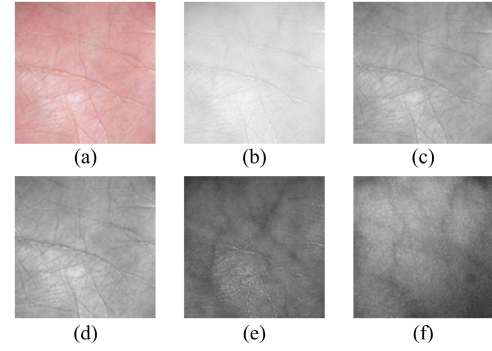


Fig. 1. Different channels of a palm image. (a) RGB. (b) Red. (c) Green. (d) Blue. (e) Saturation. (f) NIR.

Algorithm 2: Extracting palm vein ROI

Input: Hand RGB image I ;
Output: palm ROI saturation image I_{sat} ;
 1) crop to the center square image I_c and scale down to smaller image I_s (160×160 pixels);
 2) segment hand region on I_s by applying the Otsu's and Adaptive thresholding methods;
 3) find potential valleys between the adjacent fingers by tracking changes in the distance of contour points to the corresponding convex hull edge;
 4) find key vector candidates based on the bisector of angle formed by each two corresponding valley edges;
 5) select two optimal key vectors by verifying angles formed by each group of three key vector candidates;
 6) specify hand direction and scale, locate four vertices of palm ROI in I_s and scale up to coordinates in I_c ;
 7) extract palm ROI I_r from I_c using the four points-perspective transformation method;
 8) extract saturation channel I_{sat} of I_r using

Algorithm 1.

published articles address palm vein recognition systems on mobile phones using RGB images.

III. PROPOSED METHODOLOGY

This section describes the methodology of the proposed method. First, the reason for choosing the saturation channel as an input for the proposed model and saturation channel extraction are presented. Next, an improved palm ROI extraction method is described. We also present a new architecture for palm vein feature extraction in the latter part.

A. Palm Vein on Saturation Channel

The vein remains one of the most reliable biometrics for study and uses in wider applications. For example, some European projects focused on several vein programs, including security applications using NIR vein imaging with conventional computing platforms to perform human identification [29], protecting the border by checking travelers' biometrics including vein, voice, etc. [30]. In an attempt to extract veins from RGB images,

¹[Online]. Available: <https://github.com/trungvdhp/mpsnet>

[3] showed that the red channel captures palmprint information and some palm vein structures; therefore, the red channel is stretched before combining with the green and blue channels to emphasize the vein line in the RGB images. In contrast, our study detects palm vein patterns using the saturation channel, which provides high contrast imaging between the structure of palm vein lines and other structures in the background. In fact, researchers proved the saturation channel useful in several applications, such as detecting artifacts in retinal images [24] or Plasmodium segmentation [25]. Experiments indicate that the palm vein information is more evident in the saturation channel [Fig. 1(e)] than in the red channel [Fig. 1(b)], green channel [Fig. 1(c)], and blue channel [Fig. 1(d)] of the RGB image [Fig. 1(a)]. The saturation channel is also close to the NIR spectrum [Fig. 1(f)]. Algorithm 1 presents the saturation channel extraction. The algorithm converts the R, G, and B values of an RGB image to the floating-point format and scales to fit a range between 0 and 1; next, it calculates the saturation value S of the HSL space. The output of S contains the palm vein information, which the proposed system then uses. The effectiveness of this selection channel compared with the red channel was verified, as shown in Table IV.

B. ROI Extraction

This section presents an improved ROI extraction method for RGB palm images, using the convex hull and key vectors. Algorithm 2 describes the steps of the proposed method. Our previous ROI extraction method is presented in [5]. Steps 1, 5, and 7 are three improvements to complete the earlier version. The modifications make our improved algorithm more effective in a real-time environment, even in the aforementioned problem situations (e.g., large rotations, partial hand image capturing, closed fingers, or rings). Details of the algorithm are described below.

1) Cropping and Scaling: The center square image I_c is cropped from the original RGB image I and scaled down to a smaller image I_s to reduce computation time; however, this step was not used in our previous work. Note that I_s should not be too small to avoid losing the smoothness of the hand contour, separating the hand region from the complex background. In our proposed system, the size of I_s was set to 160×160 pixels.

2) Segmentation: The hand region is segmented from the background. Based on the distribution of values in various color channels from our self-collected RGB palm databases, we detected that the distribution of channels Cr and Cb of the YCrCb color space and channels A and B of the LAB color space are much more compact than others. If the corresponding number of pixels in the hand region is the largest but not larger than the threshold t , where $t = 0.8 \times$ the number of pixels in the whole image to avoid over-and under-segmentation, then that channel is selected as the most compact distribution (MC). Regarding separated fingers [Fig. 2(a)], applying Otsu's thresholding method to the MC channel also segments the hand region from the background [Fig. 2(d)]. However, when the fingers are closed [Fig. 2(b)], this process unintentionally generates another background region between the closed fingers. In this

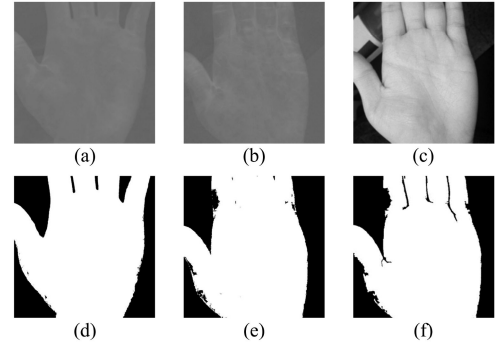


Fig. 2. Hand segmentation. (a) The most compact channel (MC channel) when open fingers. (b) MC channel when closed fingers. (c) Lightness channel. (d) Otsu's thresholding result on MC channel with open fingers. (e) Otsu's thresholding result on MC channel with closed fingers. (f) Adaptive thresholding result on lightness channel in region masked by (e).

case, Otsu's thresholding method does not separate well on the boundary between the closed fingers [Fig. 2(e)]. This may distort the ROI extraction steps. To address this problem, we use the result of Otsu's thresholding method on the MC channel as a hand region mask. The adaptive thresholding method is performed on the lightness channel region, corresponding to that mask. Adaptive thresholding provides a better binary mask [Fig. 2(f)] because the lightness channel [Fig. 2(c)] shows a higher contrast between the closed fingers and the background.

3) Finding Potential Valleys: First, the hand contour was extracted from the segmented image. Subsequently, a convex hull was discovered. One valley point exists on the hand contour in the region generated by the two convex hull points. If a whole hand image is captured, the valley point can be easily determined. However, more than one valley point exists between these two convex hull points if a hand image is captured partially, some fingers are missing, or there is a ring on a finger. The proposed correction method for the convex hull simply tracks the changes in the distance of the contour points to the corresponding convex hull edge. Some empirical rules ignore convex hull edges that do not contain potential valley points between the fingers. On the hand contour, if a contour point has another point within a five-point range, backward and forward, then these are neighbor points. The two vectors' angles are created from each contour point to its backward and forward points. The contour points with angles greater than the threshold (170°) were ignored for tracking, reducing the computing time. The length of a convex hull edge and the farthest distance from the contour point to the nearest convex hull edge must be in the range of [2, 80] pixels. As illustrated in Fig. 3, four red lines represent four retained convex hull edges containing potential valleys between adjacent fingers.

4) Finding Key Vector Candidates: This novel approach uses key vectors to specify the palm ROI without reference points used in previous studies, such as wrist or center points [13]. Note that the farthest point to the convex hull edge in each valley is not always the exact key point needed. In Fig. 3, the farthest point F obtained after correcting the convex hull deviates significantly from the desired key point K . To specify point K , two valley

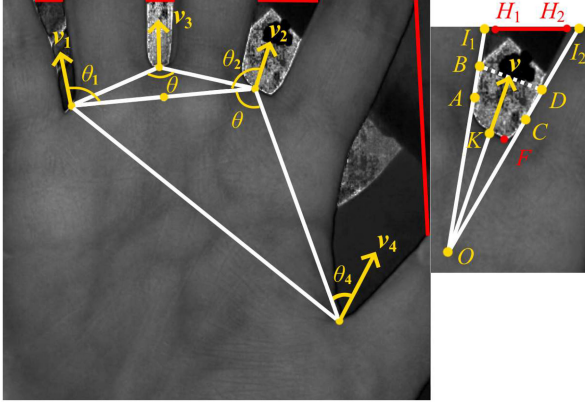


Fig. 3. Finding key vector candidates and optimal key vectors.

edges AB and CD , which are on each side of point F on the hand contour segment and are limited by two corresponding convex hull vertices H_1 and H_2 , must be detected. Two points, A and B must satisfy the following conditions: 1) Point A is far away from point B with a certain number of contour points, equal to one-third of the number of contour points from point H_1 to point F . 2) The angle of point B on the contour is greater than the threshold (170°); importantly, this obtains the contour segment between A and B , which is almost a straight line. 3) Line AB intersects with the convex hull edge H_1H_2 at point I_1 , and the distance from I_1 to H_1 must be smaller than five pixels. Similar conditions are applied to points C and D . Two valley edges AB and CD intersect at point O . The key vector candidate v lies on the bisector of the angle $\hat{O}$. The tail of vector v is the key point K , which is the intersection of this bisector with the hand contour. After obtaining all key vector candidates, the top four candidates with the longest distance from the valley point to the corresponding convex hull edge were retained.

5) Selecting Two Optimal Key Vectors: Fig. 3 illustrates key vector candidates and optimal key vectors. One key vector between the little and ring fingers and one between the index and middle fingers was selected among the candidates. If only one candidate was identified, the ROI cannot be located. If two candidates were identified, they were the two optimal key vectors. Otherwise, the candidates were divided into groups of three. The tails of the three vectors in each group form a triangle. Then, these key vector groups were sorted in descending order by the largest angle of the triangle in each group. For each sorted group, if two optimal key vectors exist, then stop. The conditions for the two optimal key vectors selected in the group are as follows: If the largest angle θ at the tail of vector v_3 is larger than or equal to a threshold angle (120°), the two remaining vectors v_1 and v_2 are selected as long as the shortest edge is longer than half of the longest edge in the corresponding triangle, and the possible ROI is inside the hand region. Conversely, assuming that the key vector between the middle-ring fingers v_3 is lost, then the largest angle θ at the tail of vector v_2 is smaller than 120° , the method of selecting two optimal key vectors for this case is as follows: If the angle formed by a vector v_4 (thumb-index valley) with tails of the two remaining vectors v_1 and v_2 is the

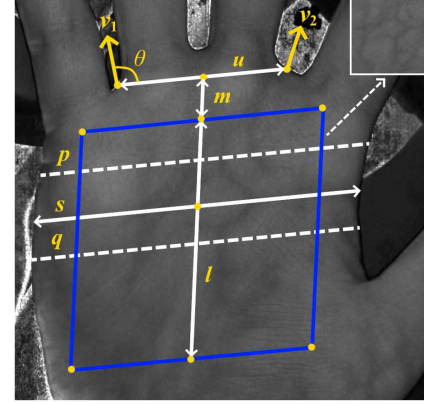


Fig. 4. Locating and extracting palm ROI.

smallest (θ_4 compared to θ_1 and θ_2), then v_4 is eliminated. The two remaining vectors are optimal if the possible ROI is inside the hand region. The above conditions can resolve the failed detection in our earlier work [5] when the algorithm cannot recognize the key vector between middle-ring fingers or when the largest angle is smaller than 120° .

6) Locating Palm ROI: Fig. 4 illustrates this step. First, determining whether a key vector lies to the left or the right of another indicates the hand's direction. Line u connects two corresponding key points of the key vectors v_1 and v_2 . The key vector v_1 is to the left of v_2 if the angle θ scanning from v_1 to line u is negative (clockwise); otherwise, v_1 is to the right of v_2 . Next, the following determines the scale of the hand: Line p is parallel to line u and the distance from p to u is half that of line u . Line q is parallel to line p and the distance from line p to q is half of line u . Similar to the approach in [22], the hand scale s is the shortest parallel line to line p and does not exceed line q . The side length of the ROI is computed using (1)

$$l = s \times (1 - 2\alpha) \quad (1)$$

where $0 < \alpha < 1$. To eliminate an ROI that is too small or too large, the size length of the ROI needs to follow the condition $h \times \sigma_1 \leq l \leq h \times \sigma_2$, where h is the height of the binary image and $0 < \sigma_1, \sigma_2 < 1$. On the line passing through the midpoint of line u and the midpoint of hand scale s , the midpoint of the ROI top edge and the midpoint of the ROI bottom edge are located. The distance m from the midpoint of the ROI top edge to the midpoint of line u is calculated by (2)

$$m = s \times \beta \quad (2)$$

where $0 < \beta < 1$. In our system, $\alpha = \beta = 0.13$, $\sigma_1 = 0.3$, $\sigma_2 = 0.8$, $h = 160$. Based on these points, we located four vertices of the ROI in I_s . If any vertex of the ROI is out of the hand region, then it is eliminated. The coordinates of the four ROI vertices in the binary image were scaled up to extract the coordinates in the cropped RGB image I_c .

7) Extracting Palm ROI: Before cropping the ROI, our earlier algorithm [5] rotated the hand image on the 2-D plane by the rotation angle θ obtained from two optimal key vectors. However, this method may not perform well when the hand rotates in a 3-D space. To resolve this, the four points-perspective

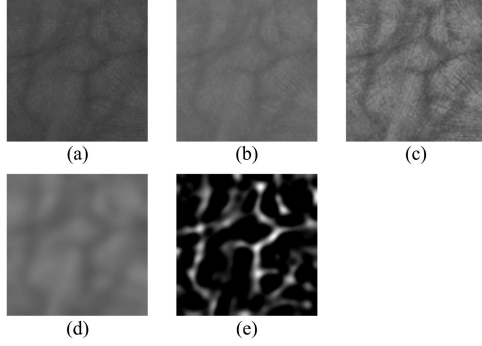


Fig. 5. Processes of palm vein enhancement. (a) Original ROI image. (b) Result of applying AGC. (c) Result of applying CLAHE on (b). (d) Result of applying low-pass filter on (c). (e) Result of applying high-pass filter on (d).

transformation function is applied to obtain an RGB palm ROI with a size of 128×128 pixels.

8) *Extracting Saturation Channel*: Algorithm 1 is used to extract the saturation channel of the RGB palm ROI.

C. Palm Vein Enhancement

To emphasize the details of the vein patterns in the palm ROI image obtained from the previous step, we use the approach based on local-ridge-enhancement (LRE) in [14]. The illumination of the ROI image shown in Fig. 5(a) is normalized using the automatic gamma correction (AGC) [15] based on the average brightness and the result is shown in Fig. 5(b). Then, the contrast of the image is enhanced, using the contrast limited adaptive histogram equalization (CLAHE) [16] with a clip-limit value of 2 and a grid size of 4×4 and Fig. 5(c) shows the result. The image is then normalized to a range of 0 to 1.0. Next, a Gaussian low-pass filter with a standard deviation of 4 removes the noise and the result is shown in Fig. 5(d). Finally, the Laplacian filter acts as a high-pass filter to acquire the vein details, as shown in Fig. 5(e). The pixel values of the final image were truncated to zero if the pixel values were smaller than 0. The image was normalized to fit from 0 to 255.

D. Proposed Architecture of Feature Extractor: MPSNet

As mentioned earlier, in the past, the deep learning-based methods for palm vein recognition enhanced the architectures and used different loss functions to increase the recognition accuracy. PalmR_{CNN} [6] modified scalable “Inception_ResNet_v1” and joined the center loss with the softmax loss to make the features learned by the network more representative. However, this loss results in drastic memory consumption with increased CNN layers. DHPN [7] modified the CNN-F architecture, obtaining a binary code to match and recognize palm vein images. Obviously, most of the conventional architectures stacked traditional convolution layers to enhance the recognition rate; thus, they are large and unsuitable for smartphones with resource limitations. This article proposes a new lightweight deep learning model, MPSNet shown in Fig. 6, for palm vein feature extraction, using the saturation channel. The details

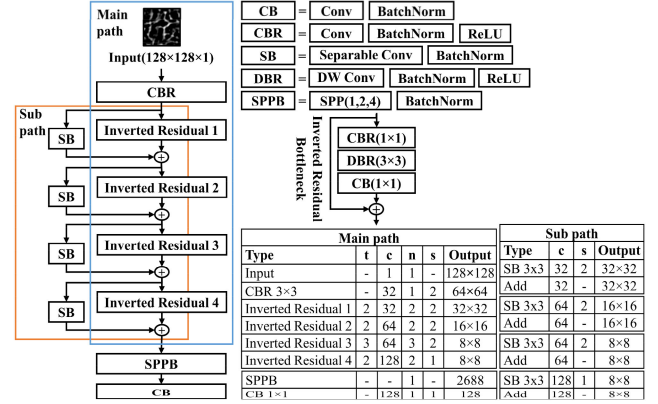


Fig. 6. MPSNet: A lightweight deep learning model for palm vein feature extraction using saturation channel.

of the proposed model are analyzed and compared to several works specializing in mobile devices, such as MobileNetV1 [9], MobileNetV2 [17], MobileNetV3 [18], and MobileFaceNet [20]. The framework is composed of two paths (main path and sub-path), one spatial pyramid max pooling block (SPPB), and one linear convolution 1×1 block (CB) as the output of the feature embedding. The main path accepts an input of the saturation channel image, extracts features by one convolutional block (CBR block), and transmits information through four inverted residual blocks [17].

The subpath contains four linear separable convolutional blocks (SBs). It receives the output of the CBR and passes it to the first SB. Then, the first SB’s output is added with the output of Inverted Residual 1 and passes the result to the second SB; then, these operations repeat three times, before passing the result to SPPB. This enhances the feature representations and prevents gradient vanishing during the training phase. Note that the previous layer’s output is passed through the SB to synchronize the feature map dimension for the Add layer. Using a depthwise separable convolutional layer in most of the added connections not only saves a large number of parameters but also keeps learning palm vein features compared to the traditional convolutional layer. The details of the CBR, SB, and Inverted Residual block are shown in Fig. 6. In fact, summing up the output of the fourth SB and the output of Inverted Residual 4 is to fuse the main path and the sub-path completely. The SPPB receives a feature map after the main path and the sub-path are fused together, then pools and transfers to one linear convolutional layer to output the expected feature embedding. In this study, the proposed model accepts an input of a saturation channel of size $128 \times 128 \times 1$ and outputs a feature map size of 128. An Inverted Residual block consists of several repetitions of the bottleneck module, which is composed of one CBR (1×1) to extend the channel, one DBR (3×3) to extract latent features, and one CB (1×1) to reduce the feature map’s channel. The expansion factor t , the number of filters or channels c , strides s , and repeat n of the bottle next module are set up, as shown in Fig. 6 after considering trade-offs of accuracy and computational complexity.

In our proposed architecture, feature fusion consists of three elements—connections in the sub-path, aggregation between two paths, and the SPP module. Feature fusion enhances discriminative features and increases the accuracy of the overall recognition process more than versions without it. Connections in the sub-path and aggregation between two paths are set up to prevent transmission congestion or gradient vanishing in a deep neural network. Fig. 6 shows that, after the CBR block, the computations are divided into two paths and then fused together. Equation (3) defines the fusion rule

$$f_l = x_l + x_{l-1} \quad (3)$$

where f_l is the fused feature map at l^{th} layer, x_l is the feature map of l^{th} layer, x_{l-1} is the feature map of $(l-1)^{th}$ layer. Note that the output of $(l-1)^{th}$ layer is passed through the linear SB to synchronize the feature map dimensions for the Add operation. The fused feature map is created after summing up the output of the fourth SB, and the output of the inverted residual 4.

Previously, using the global average pooling layer in MobileNetV1 [9], MobileNetV2 [17], MobileNetV3 [18], PalmRCNN [6], or linear global depthwise convolutional layer in MobileFaceNet [20], or direct transfer of the convolutional layer output to the fully connected layer in DHPN [7] could be biased to global features. Thus, our model uses the Spatial Pyramid Pooling module [19] to obtain local and global features after the fused feature map. SPP uses the fused feature map of the main path and sub-path as the input and generates a feature vector by pooling the input feature map into several predefined bins. Instead of using the average pooling of each bin, we selected max pooling to avoid local noise generated unintentionally. The responses of all bins are then concatenated to a fixed feature vector of size $k \times B$, where B is the number of bins and k is the number of channels of the fused feature map. In our proposed architecture, for the input fused feature map of size 8×8 with 128 channels, we created twenty-one (21) bins using three (3) pooling scales, 1×1 , 2×2 , and 4×4 , respectively, to generate the feature vector. By using different scales of pooling sizes, the SPP aggregates the features at the local and global levels. Treating the local feature as important as the global feature enhances the overall accuracy of the palm vein recognition model. Finally, SPP generates a feature vector of size $21 \times 128 = 2688$.

IV. EXPERIMENTAL RESULTS

In this section, extensive experiments are performed to evaluate the biometric performance of MPSNet on databases, as well as the real-time performance on smartphones.

A. Databases

Previous studies on palm vein recognition used NIR databases, such as PolyU [11], CASIA [10], and Tongji [22], obtained by self-built devices using Charge-coupled Device (CCD) cameras and LED light sources. The large-scale Tongji Contactless Palm vein database consists of 12 000 images from 300 volunteers using their device with a ring of 940 nm LED light source in two sessions in a partly constrained environment.

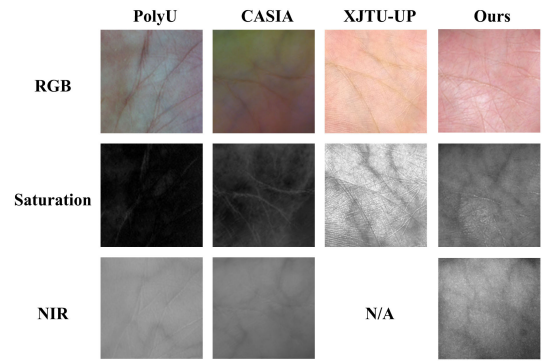


Fig. 7. Comparison of channels between different palm databases: PolyU and CASIA are collected by their self-built devices using CCD camera and LED light sources while XJTU-UP and our databases are collected by smartphones.

For the PolyU [11] database, each palm was captured under four spectra of 470 nm (blue), 525 nm (green), 660 nm (red), and 880 nm (NIR) in an ideal laboratory environment. Each spectrum captured 6000 images from 250 individuals in two sessions. For CASIA [10], each palm was captured under six spectra of 460, 630, 700, 850, and 940 nm. This database contains 1200 images of each spectrum from 100 individuals in two sessions. Although collectors aim to increase the diversity of images by allowing a certain degree of variation in hand posture, the controlled illumination is far from practical.

Note that PolyU and CASIA databases also provided images of red, green, and blue channels, which were acquired separately by switching quickly to the corresponding illuminator. Hence, they did not precisely represent the intensity of the color in the actual RGB hand image. In Fig. 7, the reconstructed RGB images from the PolyU and CASIA databases do not show the natural color of the palm skin. This leads to the generation of saturation images from PolyU and CASIA, not including sufficient palm vein for identification, compared to those from the actual RGB images in Xian Jiaotong University Unconstrained Palmprint (XJTU-UP) [12] and our self-collected databases. Therefore, PolyU and CASIA only provided NIR images in our experiments. To evaluate the biometric performance of the saturation channel, we used the RGB palm databases, as described below. Our experiments used 1960 RGB images captured by an iPhone 6S with LED flashlight from the XJTU-UP database. Similar to the aforementioned databases, XJTU-UP was captured indoors and in a controlled environment. To further additional research, we provide our self-collected database, consisting of 2180 RGB images from 109 people captured by HTC One A9 (NTUST-HP) and 5760 RGB images from 144 people by iPhone 8+ (NTUST-IP) in an unconstrained manner with the flashlight. Fig. 8 shows some samples from our database. Our database's contactless RGB palm vein images were collected from volunteers from China, Taiwan, Vietnam, Thailand, Indonesia, and Mongolia aged from 18 to 63 years old with various backgrounds, rotations, scales, and different illuminations. The NTUST-HP database was captured in one session, with ten samples indoors for each palm. NTUST-IP was captured in two sessions, ten samples for each palm indoor, and ten samples for



Fig. 8. Several hand samples from our self-collected databases.

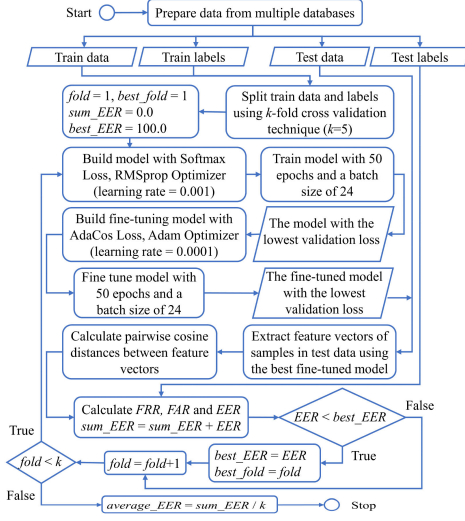


Fig. 9. Flowchart for training and testing algorithms.

each palm outdoors. The time interval between the two sessions was about one week. Obviously, our self-collected database is highly variant and is captured in both indoor and outdoor environments. The number of individuals in our databases is 253, which is less than that of Tongji, but larger than others. We used the ROI palm image provided by PolyU and XJTU-UP and extracted the ROI for other databases using our proposed ROI extraction algorithm in Section III.B.

B. Experiment Settings

Several experiments were conducted to evaluate MPSNet performance, the proposed model from Section III-D. The experimental results are compared with other related works, such as PalmR_{CNN} [6], DHPN [7], PVSNet [8], and some mobile architectures such as MobileNetV1 [9], MobileNetV2 [17], MobileNetV3 [18], and MobileFaceNet [20]. For PalmR_{CNN} [6], DHPN [7], and PVSNet [8]; as such, the results used for the comparison are reported in their corresponding papers. All the experiments in this study were trained and tested using the same settings for a fair comparison. Fig. 9 illustrates the flowchart of the training and testing algorithms. The details are as follows:

1) **Training Setting:** All models—including MPSNet, MobileNetV1 [9], MobileNetV2 [17], MobileNetV3 [18], and MobileFaceNet [20]—were trained using palm ROI images from multiple databases listed in the previous section and using

the same desktop environment with Python 3.6.8, Keras-GPU 2.2.4 with Tensorflow-GPU backend, and NVIDIA GTX 1080 GPU. All list models are first pre-trained with softmax loss and fine-tuned with AdaCos [21], one by one. All layers before the feature embedding layer (linear convolutional layer) were frozen in the fine-tuning step. The batch size of the training process was 24. The “He uniform” kernel initialization [23] and the L2 kernel regularization with weight decay of 0.0005 are used for all layers. In the pre-trained classification model with softmax loss, the optimizer function was RMSprop, the learning rate was 0.001, and the number of epochs was 50. In the fine-tuning step, the AdaCos loss function was imposed, the optimizer function was Adam, the learning rate was lower at 0.0001 to avoid a negative effect on the learned features, and the number of epochs was 50. AdaCos maximize the angular margins between different identities to achieve optimal generalization. The benefit of AdaCos over other cosine-based softmax losses is that it can produce reasonable scale parameters by sensing the training convergence of the model in the current iteration. The results in Table I indicate that AdaCos is better than the other losses on all Open-sets. The trained model, which had the lowest validation loss, was retained. The model was evaluated using the k -fold cross-validation technique ($k = 5$).

2) **Testing Setting:** The testing phase consisted of two scenarios: (1) Closed-set (CS): the test set is included in the training set; and (2) Open-set (OS): the test set is not included in the training set. The trained model is used as the feature extractor, which produces a feature vector of 128 features for each input test image. The weights and parameters were loaded during the training phase. The verification process compared two specified feature vectors to verify whether they were from the same palm. Suppose that there is a set of palms, $P = \{P_i, i = 1 \dots n\}$. Let v_i and v_j be the feature vectors of palms P_i and P_j , respectively. The number of features of each vector is m , which is equal to 128. The distance between v_i and v_j is calculated using the cosine distance, as shown in (4).

$$d_{i,j} = 1 - \frac{v_i \cdot v_j}{\|v_i\| \|v_j\|} = 1 - \frac{\sum_{k=1}^m v_{ik} v_{jk}}{\sqrt{\sum_{k=1}^m v_{ik}^2} \sqrt{\sum_{k=1}^m v_{jk}^2}}. \quad (4)$$

d_t be the cosine distance threshold, v_i and v_j are from the same palm if $d_{i,j} \leq d_t$; otherwise, they represent different palms. The matching distance distribution is generated based on distances between all pairs of feature vectors in the test set. Changing the threshold d_t determines False Rejected Rate (FRR), False Accepted Rate (FAR), the EER, and detection error tradeoff (DET) curves.

C. Results

1) **Biometric Performance on NIR Databases:** The number of parameters and model sizes are listed in Table II. Table III shows the EERs among the models on different NIR test sets. The MPSNet model utilizes 2.54 MB of memory with 0.59M parameters, taking much less space than other models. MobileFaceNet possesses the second-best EER for both closed and open sets, respectively. Compared to the second-best model, for the closed-set Tongji, PolyU, and CASIA, the MPSNet model

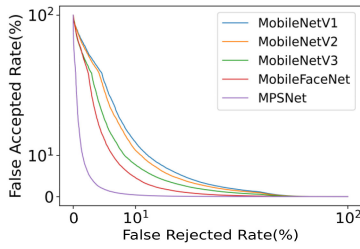


Fig. 10. DET Curve of Models Tested on Open-Set NTUST-IP (two sessions).

EERs are 0.0001%, 0.0011%, and 0.17%, and which are reduced by 15, 4.1, and 2.4 times, respectively. Similarly, for the open-set Tongji, PolyU, and CASIA, the EERs of the MPSNet model are 0.46%, 0.47%, and 1.52%, which are reduced by 1.8, 1.6, and 1.7 times, respectively. The experiment indicated that combining saturation images and NIR images for the training process is possible, while achieving high palm vein identification results on NIR data.

2) Biometric Performance on RGB Databases: The experiment was conducted on open-set RGB databases, captured by smartphones to compare 1) the biometric performance among models and 2) between the palm vein features of the saturation channel and the red channel, as shown in Table IV. Because PalmR_{CNN} [6], DHPN [7], and PVSNet [8] did not provide the code publicly and the results on RGB databases were not reported, a comparison between using the saturation and the red channels for palm vein recognition could only be performed on MobileNet architectures [9], [17], [18], [20] and our proposed MPSNet. Experimental results indicate that our MPSNet has the best EER compared to the other models under the same conditions. Specifically, for the test datasets XJTU-UP, NTUST-HP, and NTUST-IP, the EERs of MPSNet are 0.90%, 0.49%, and 3.62%, which are reduced by 1.4, 1.1, and 2.2 times compared to the second-best model, MobileFaceNet, respectively. In addition, the saturation channel provides better biometric performance than the red channel. When using saturation channels, the EERs of our MPSNet model on one-session test sets of XJTU-UP and NTUST-HP were 16% and 10% lower than those using the red channel, respectively. In the two-session test set of NTUST-IP, the EER of our MPSNet model using the saturation channel was 22% lower than that of the red channel. The experimental results indicate that using the saturation channel is more efficient than using the red channel for palm vein verification on RGB palm images.

For each method, the model with the best EER for each test database was used to draw the DET curve. Fig. 10 illustrates that the DET curve of our MPSNet model is lower than those of the other models when testing on our database NTUST-IP (two sessions), which indicates our proposed model's effectiveness. The matching distance distributions of genuine and impostor for the MPSNet model on the NTUST-HP and NTUST-IP databases are illustrated in Fig. 11. Because the EER for one session is significantly smaller than those for two sessions, the interference between the genuine distances and impostor distances for one

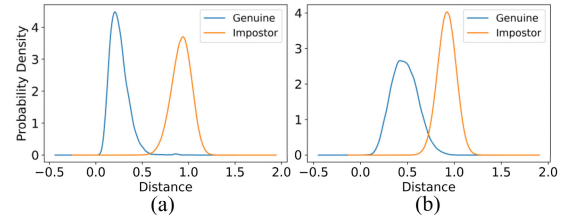


Fig. 11. Matching distance distribution of MPSNet model with best EER on our (a) NTUST-HP and (b) NTUST-IP test databases.

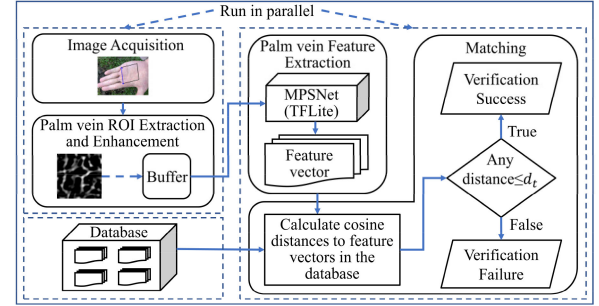


Fig. 12. Real-time palm vein recognition systems on smartphones.

session shown in Fig. 11(a) is smaller than that for two sessions shown in Fig. 11(b).

3) Real-Time Results: In this experiment, we designed real-time palm vein recognition applications for two types of smartphones. Fig. 12 illustrates the procedure for the proposed system. The regular back camera of a smartphone captures an individual's hand while the LED flashlight is turned on. Our designed system selects a camera resolution of 640×480 pixels to balance the trade-off between image quality and processing time. The process includes three modules—preprocessing, feature extraction, and matching. The preprocessing module consists of image acquisition, palm vein ROI extraction, and enhancement. The preprocessing module was designed to run in parallel with other modules. Once a frame finishes preprocessing, the palm vein data are transferred to the buffer, and the next frame is handled instantly. The feature extraction module receives the palm vein data ready in this buffer, and then extracts and sends the feature vector to the matching module. This module compares the cosine distance of the authenticated feature vector with those stored in the database and decides whether access is granted. To design applications for two kinds of smartphones of Android and iOS operating systems, we used Android Studio IDE with Kotlin programming language and XCode IDE with Objective-C programming language, respectively. Then, we install and run the applications on an Android device (HTC One A9) and an iOS device (iPhone 8+) to measure the performance of the models. As mentioned in the previous steps, PalmR_{CNN} [6], DHPN [7], and PVSNet [8] models were not used for this experiment because they were not designed for smartphones. Our proposed MPSNet and other MobileNet models with the best performance on two databases, NTUST-HP and NTUST-IP, were used in applications.

TABLE V
REAL-TIME TESTING RESULTS (SINGLE CPU THREAD)

			HTC One A9		iPhone 8+	
Preprocessing Time (ms)			85		14	
Methods	MSize (MB)	MFLOPs	FET (ms)	RS (fps)	FET (ms)	RS (fps)
MobileNetV1 [9]	3.22	70	120	5	15	58
MobileNetV2 [17]	2.33	69	119	5	16	57
MobileNetV3 [18]	3.02	63	109	5	13	59
MobileFaceNet [20]	1.04	162	280	3	47	19
MPSNet [Ours]	0.62	37	64	9	8	60

Note: MSize = TFLite Model Size; FET = Feature Extraction Time; RS = Real-Time Speed.

Table V shows the measured results for two different smartphones. The models are converted to Tensor Flow Lite (TFLite) format, which is smaller and can run inferences faster on smartphones. In terms of the preprocessing time and feature extraction time, the computation time is the average time required to handle one frame within three seconds of the recognition process. The real-time speed is measured by the number of frames in which the system can successfully handle all tasks within one second. One CPU thread was used for feature extraction. The results show that our proposed MPSNet model is the best among the compared models. The TFLite model size of our MPSNet model is the smallest at 0.62 MB. The number of floating-point operations performed by our MPSNet model during a single inference is approximately 37 MFLOPs, 41% smaller than that of the second smallest-MobileNetV3. For feature extraction time, on HTC One A9, the result of MPSNet is 64 ms, which is 337% and 70% faster than that of MobileFaceNet and MobileNetV3. For iPhone 8+, the rate is 8 ms, which is 487% and 62% faster than those of MobileFaceNet and MobileNetV3. For the real-time speed, on HTC One A9, the rate of MPSNet is 9 fps, which is 3 and 1.5 times more frames than those of MobileFaceNet and MobileNetV3, respectively. On iPhone 8+, it is 60 fps, which is 216% and approximately 1% higher than MobileFaceNet and MobileNetV3, respectively.

V. CONCLUSION

In this article, a novel lightweight, real-time, contactless palm vein recognition system (MPSNet) was developed for smartphones. The system uses a regular back camera and LED flash-light mounted on smartphones to capture and recognize palm images. Experimentation proved that the system is efficient in terms of economy and portability. The improved palm vein ROI extraction method proposed herein effectively copes with scale, rotation, translation, and non-uniform illumination variation. In addition, extracting palm vein features from the saturation channels provides higher accuracy than from the red channel alone. Results show that the new proposed model for feature extraction achieves higher accuracy than related models and is smaller. In particular, our current MPSNet achieves the best EER of 0.49% and an inference time of 8 ms. The palm vein recognition system for smartphones can be used to manage access to physical and digital resources, such as buildings, rooms, computing devices, and online payment without a password. For future research, we

will study to improve the accuracy of the system, using multiple biometrics.

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Shi-Jinn Horng received the Ph.D. degree in computer science from National Tsing Hua University, Hsinchu, Taiwan, in 1989.

He is currently the Chair Professor with the Department of Computer Science and Information Engineering, National Taiwan University of Science and Technology, Taipei, Taiwan. He has authored or coauthored more than 300 papers. His research interests include biometric recognition, deep learning, and information security.



Dinh-Trung Vu received the M.S. degree in computer science from National Taiwan Ocean University, Keelung, Taiwan, in 2015. He is currently working toward the Ph.D. degree with the National Taiwan University of Science and Technology, Taipei, Taiwan, both in computer science.

His research interests include biometric recognition, image processing, and deep learning.



Thi-Van Nguyen received the M.S. degree from National Taiwan Ocean University, Keelung, Taiwan, in 2015. She is currently working toward the Ph.D. degree with the National Taiwan University of Science and Technology, Taipei, Taiwan, both in computer science.

Her research interests include biometric recognition, image processing, and deep learning.



Wanlei Zhou (Senior Member, IEEE) received the Ph.D. degree in computer science from The Australian National University, Canberra, Australia, in 1991.

He is currently the Vice Rector and Dean with the Institute of Data Science, City University of Macau, Macao SAR, China. He was the Head with the School of Computer Science, University of Technology Sydney (UTS), Ultimo, Australia. He has authored or coauthored more than 400 papers. His research interests include security and privacy, bioinformatics, and e-learning.

Dr. Zhou was the recipient of more than ten ARC grants.



Chin-Teng Lin (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Purdue University, West Lafayette, IN, USA, in 1992.

He is currently the Distinguished Professor with the Faculty of Engineering and Information Technology, and the Co-Director of Center for Artificial Intelligence, University of Technology Sydney, Ultimo, Australia. He has authored or coauthored more than 300 journal papers in the areas of neural networks, fuzzy systems, and brain computer interface.

Dr. Lin was an IFSA Fellow in 2012.

He was the Editor-in-Chief of IEEE TFS from 2011 to 2016, and the Deputy EiC of IEEE TCS-II in 2006–2008. He was the recipient of the IEEE Fuzzy Systems Pioneer Awards in 2017.